

PVsyst Simulation Report

As-Built System Performance Analysis

Desert Bloom Solar + Storage

75 MW AC / 98.5 MW DC Photovoltaic Power Plant
with 15.6 MWh / 7.6 MW Battery Energy Storage

12500 Mojave Desert Highway, Barstow, CA 92311
Latitude: 34.8958N | Longitude: 117.0173W

PVsyst Version: 7.4.2

Report Date: March 28, 2024

Prepared by: Radiant Energy Consultants, LLC

Second Buyer Independent Engineer Report

CONFIDENTIAL - PREPARED FOR SOLARA HOLDINGS LLC

1. PROJECT SUMMARY

Project Name:	Desert Bloom Solar + Storage
Project Owner:	Desert Bloom Solar LLC (subsidiary of Solara Holdings LLC)
Location:	12500 Mojave Desert Highway, Barstow, CA 92311
County:	San Bernardino County, California
Coordinates:	34.8958N, 117.0173W Elevation: 686m
Climate Zone:	Desert (BWk) - Hot arid climate
System Size (DC):	98.5 MWp
System Size (AC):	75.0 MW
DC/AC Ratio:	1.313
Module Technology:	Bifacial monocrystalline PERC
Module Manufacturer:	LONGi Solar
Module Model:	LR5-72HBD-545M
Module Quantity:	180,734 modules
Module STC Rating:	545 Wp
Inverter Manufacturer:	Sungrow
Inverter Model:	SG250HX
Inverter Quantity:	12 units
Inverter Rated Power:	250 kW each (3,000 kW total per unit)
Tracking System:	Single-axis horizontal tracking (N-S axis)
Tracker Manufacturer:	NEXTracker NX Horizon
Tilt Range:	+/- 60 degrees
Ground Coverage Ratio:	0.33
Row Spacing:	6.5m (center-to-center)
Battery Storage:	Tesla Megapack 2XL, 4 units
Battery Capacity:	15,664 kWh total (3,916 kWh each)
Battery Power:	7,648 kW total (1,912 kW each)
Interconnection:	115 kV at Desert Bloom Switching Station
Offtaker:	Southern California Edison (SCE)
PPA Term:	20 years from COD
Commercial Operation Date:	May 1, 2024

2. METEOROLOGICAL DATA

Data Source: SolarAnywhere Enhanced Resolution Time-Series (ERTS) dataset, validated against ground-based measurements from the nearby Barstow-Daggett SURFRAD station (10.3 km from site).

Period of Record: January 2000 through December 2023 (24 years)

Spatial Resolution: 0.01 degrees (~1 km)

Temporal Resolution: Hourly

Long-term average meteorological parameters:

Month	GHI	DNI	DHI	Temp (C)	Wind (m/s)
January	118.2	169.4	33.8	9.1	3.2
February	140.6	191.2	40.1	11.3	3.5
March	192.8	254.3	52.7	14.8	3.8
April	228.4	310.6	58.2	19.2	4.1
May	262.1	348.9	63.5	24.6	4.3
June	278.6	374.2	60.8	30.1	4.0
July	271.3	352.8	66.4	34.2	3.7
August	249.8	332.1	61.9	33.1	3.5
September	210.4	296.7	51.3	28.4	3.3
October	168.3	243.8	42.6	21.3	3.1
November	124.5	183.6	34.2	13.7	3.0
December	105.8	157.2	30.4	8.8	3.1
Annual	2350.8	3214.8	595.9	20.7	3.6

Units: GHI, DNI, DHI in kWh/m2; Temperature in degrees Celsius; Wind speed in m/s at 10m height.
GHI = Global Horizontal Irradiance; DNI = Direct Normal Irradiance; DHI = Diffuse Horizontal Irradiance

3. DETAILED LOSS ANALYSIS

The following table summarizes the loss factors applied in the PVsyst simulation model, reflecting as-built system configuration and site-specific conditions:

Loss Category	Loss Factor
Near Shading (electrical)	0.8%
IAM (Incidence Angle Modifier)	2.1%
Soiling Loss	3.5%
DC Wiring Losses	1.5%
Module Quality / LID	1.0%
Module Mismatch	1.5%
Temperature Loss (annual avg)	6.2%
Inverter Efficiency Loss	1.8%
AC Wiring Loss	0.5%
MV Transformer Loss	0.8%
MV Line Loss	0.3%
Auxiliary Power Consumption	0.4%
Tracker Misalignment / Backtracking	1.2%
Bifacial Gain	-4.8% (gain)
Curtailment Estimate	0.5%
Availability Loss	1.0%
Grid Connection Loss	0.2%
TOTAL SYSTEM LOSS	16.8%
PERFORMANCE RATIO (PR)	83.2%

4. ENERGY PRODUCTION RESULTS

4.1 Annual Production Summary

Parameter	Value
P50 Annual Production	182,400 MWh
P75 Annual Production	174,200 MWh
P90 Annual Production	168,500 MWh
P95 Annual Production	164,800 MWh
P99 Annual Production	158,200 MWh
Performance Ratio (PR)	83.2%
Specific Yield	1,852 kWh/kWp
Capacity Factor	27.8%
Energy Yield (per m2)	421 kWh/m2/year
Degradation Rate	0.40% per year (years 2-25)
Year 1 Degradation	2.0% (LID + initial)

4.2 Monthly Energy Production (P50)

Month	GlobInc	GlobEff	E_Grid	PR (%)	CF (%)
January	145.2	138.4	10,180	80.1	18.5
February	168.4	160.8	12,040	81.5	24.1
March	226.8	217.2	16,420	82.5	29.5
April	264.6	253.8	18,960	83.1	35.3
May	296.4	284.8	20,840	83.6	37.4
June	310.2	298.4	21,260	83.8	39.5
July	304.8	292.6	20,460	82.4	36.7
August	282.4	271.0	19,280	82.8	34.6
September	244.6	234.8	17,400	83.2	32.3
October	198.4	190.2	14,280	83.5	25.6
November	152.6	145.8	10,820	82.0	20.1
December	128.4	122.4	9,460	81.2	17.0
TOTAL	2722.8	2610.2	191,400	83.2	27.8

Units: GlobInc, GlobEff in kWh/m2; E_Grid in MWh.
GlobInc = Global Incident on collector plane; GlobEff = Effective Global after losses; E_Grid = Energy injected into grid.
Note: Monthly totals are normalized to typical meteorological year.

5. UNCERTAINTY ANALYSIS

The uncertainty analysis quantifies the range of expected annual energy production based on variability in solar resource, model accuracy, and system performance parameters.

5.1 Sources of Uncertainty

The following uncertainty components were evaluated and combined using the root-sum-square (RSS) method:

Solar Resource Inter-Annual Variability:	4.8%
Solar Resource Data Quality:	2.5%
PVsyst Model Accuracy:	2.0%
Module Power Tolerance:	1.5%
Degradation Rate Uncertainty:	1.0%
Soiling Variability:	1.5%
Inverter Efficiency Uncertainty:	0.5%
Grid Availability / Curtailment:	1.0%
Combined RSS Uncertainty (1-sigma):	6.3%

5.2 Exceedance Probability Table

Probability	Energy (MWh)	% of P50	Revenue Impact
P50 (50%)	182,400	100.0%	\$7,752,000
P75 (75%)	174,200	95.5%	\$7,403,500
P90 (90%)	168,500	92.4%	\$7,161,250
P95 (95%)	164,800	90.3%	\$7,004,000
P99 (99%)	158,200	86.7%	\$6,723,500

Revenue impact calculated at PPA contract rate of \$42.50/MWh. Actual revenue will depend on curtailment events, grid availability, and settlement adjustments.

6. LONG-TERM DEGRADATION FORECAST

The following 25-year energy forecast incorporates module degradation per manufacturer specifications and industry-standard assumptions. Year 1 includes an initial Light-Induced Degradation (LID) adjustment of 2.0%, with subsequent years degrading at 0.40% per annum.

Year	P50 (MWh)	P90 (MWh)	Degrad.	Cum. Deg.	Revenue (P50)
1	178,752	165,130	2.00%	2.00%	\$7,596,960
2	178,037	164,469	0.40%	2.39%	\$7,566,572
3	177,325	163,812	0.40%	2.78%	\$7,536,306
4	176,616	163,156	0.40%	3.17%	\$7,506,161
5	175,909	162,504	0.40%	3.56%	\$7,476,136
6	175,205	161,854	0.40%	3.94%	\$7,446,231
7	174,505	161,206	0.40%	4.33%	\$7,416,447
8	173,807	160,561	0.40%	4.71%	\$7,386,781
9	173,111	159,919	0.40%	5.09%	\$7,357,234
10	172,419	159,280	0.40%	5.47%	\$7,327,805
11	171,729	158,642	0.40%	5.85%	\$7,298,493
12	171,042	158,008	0.40%	6.23%	\$7,269,299
13	170,358	157,376	0.40%	6.60%	\$7,240,222
14	169,677	156,746	0.40%	6.98%	\$7,211,261
15	168,998	156,119	0.40%	7.35%	\$7,182,416
16	168,322	155,495	0.40%	7.72%	\$7,153,687
17	167,649	154,873	0.40%	8.09%	\$7,125,072
18	166,978	154,253	0.40%	8.45%	\$7,096,572
19	166,310	153,636	0.40%	8.82%	\$7,068,185
20	165,645	153,022	0.40%	9.19%	\$7,039,913
21	164,982	152,410	0.40%	9.55%	\$7,011,753
22	164,322	151,800	0.40%	9.91%	\$6,983,706
23	163,665	151,193	0.40%	10.27%	\$6,955,771
24	163,011	150,588	0.40%	10.63%	\$6,927,948
25	162,359	149,986	0.40%	10.99%	\$6,900,236

7. CONCLUSIONS AND RECOMMENDATIONS

7.1 Key Findings

Based on our independent review of the as-built system configuration, site-specific meteorological data, and PVsyst simulation results, Radiant Energy Consultants concludes the following:

1. The P50 annual energy production estimate of 182,400 MWh is reasonable and consistent with the site's solar resource and system design. This represents a specific yield of 1,852 kWh/kWp, which is within the expected range for a single-axis tracking system in the Mojave Desert region.

2. The Performance Ratio (PR) of 83.2% is appropriate for a utility-scale bifacial tracking system in a hot-arid climate, accounting for temperature-related losses and the beneficial impact of bifacial energy gain.

3. The P90 estimate of 168,500 MWh provides adequate coverage for debt service requirements under the permanent financing facility with KeyBank National Association, maintaining a minimum DSCR of 1.25x at the P90 production level.

4. The battery energy storage system (15.6 MWh / 7.6 MW) is not modeled as a production contributor in this analysis but provides ancillary services revenue and peak-shifting capabilities that enhance overall project economics.

5. The assumed degradation rate of 0.40% per year (after Year 1 LID) is conservative relative to LONGi's 25-year linear warranty guarantee of 0.40%, providing additional margin in long-term production forecasts.

7.2 Recommendations

1. Continue monitoring system performance against the P50 baseline using the Also Energy DAS platform, with particular attention to soiling losses during the dry season (May-October).

2. Implement a structured soiling study program to validate the 3.5% soiling loss assumption and optimize cleaning schedules.

3. Review tracker backtracking algorithms annually to minimize inter-row shading losses while maximizing bifacial gain.

4. Consider updating the PVsyst model after the first full year of operation (May 2025) to calibrate simulation parameters against actual measured performance.

7.3 Certification

This report was prepared by Radiant Energy Consultants, LLC as the Second Buyer Independent Engineer for the Desert Bloom Solar + Storage project. The analysis was conducted in accordance with industry best practices and ASTM E2848 standards for photovoltaic system performance estimation.

Prepared by:

Dr. Sarah M. Rodriguez, P.E.

Senior Vice President, Renewable Energy Advisory

Radiant Energy Consultants, LLC

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